A Technical Introduction to Mixture model

Section 1

K , N = as above φ i = 1 … K , φ = as above z i = 1 … N , x i = 1 … N = as above θ i = 1 … K = { μ i = 1 … K , σ i = 1 … K 2 } μ i = 1 … K = mean of component i σ i = 1 … K 2 = variance of component i μ 0 , λ , ν , σ 0 2 = shared hyperparameters μ i = 1 … K ~ N (μ 0 , λ σ i 2) σ i = 1 … K 2 ~ I n v e r s e - G a m m a ■ (ν , σ 0 2) φ ~ S y m m e t r i c - D i r i c h l e t K ■ (β) z i = 1 … N ~ Categorical ■ (φ) x i = 1 … N ~ N (μ z i , σ z i 2) {\displaystyle {\begin{array}{lcl}K,N&=&{\text{as above}}\\\phi _{i=1\dots K},{\boldsymbol {\phi }}&=&{\text{as above}}\\z_{i=1\dots N},x_{i=1\dots N}&=&{\text{as above}}\\\theta _{i=1\dots K}&=&\{\mu _{i=1\dots K},\sigma _{i=1\dots K}^{2}\}\\\mu _{i=1\dots K}&=&{\text{mean of component }}i\\\sigma _{i=1\dots K}^{2}&=&{\text{variance of component }}i\\\mu _{0},\lambda ,\nu ,\sigma _{0}^{2}&=&{\text{shared hyperparameters}}\\\mu _{i=1\dots K}&\sim &{\mathcal {N}}(\mu _{0},\lambda \sigma _{i}^{2})\\\sigma _{i=1\dots K}^{2}&\sim &\operatorname {Inverse-Gamma} (\nu ,\sigma _{0}^{2})\\{\boldsymbol {\phi }}&\sim &\operatorname {Symmetric-Dirichlet} _{K}(\beta)\\z_{i=1\dots N}&\sim &\operatorname {Categorical} ({\boldsymbol {\phi }})\\x_{i=1\dots N}&\sim &{\mathcal {N}}(\mu _{z_{i}},\sigma _{z_{i}}^{2})\end{array}}}

Section 2

Other methods Some of them can even probably learn mixtures of heavy-tailed distributions including those with infinite variance (see links to papers below). In this setting, EM based methods would not work, since the Expectation step would diverge due to presence of outliers.

Section 3

Moment matching The method of moment matching is one of the oldest techniques for determining the mixture parameters dating back to Karl Pearson's seminal work of 1894. In this approach the parameters of the mixture are determined such that the composite distribution has moments matching some given value. In many instances extraction of solutions to the moment equations may present non-trivial algebraic or computational problems. Moreover, numerical analysis by Day has indicated that such methods may be inefficient compared to EM. Nonetheless, there has been renewed interest in this method, e.g., Craigmile and Titterington (1998) and Wang. McWilliam and Loh (2009) consider the characterisation of a hyper-cuboid normal mixture copula in large dimensional systems for which EM would be computationally prohibitive. Here a pattern analysis routine is used to generate multivariate tail-dependencies consistent with a set of univariate and (in some sense) bivariate moments. The performance of this method is then evaluated using equity log-return data with Kolmogorov–Smirnov test statistics suggesting a good descriptive fit.

Section 4

The maximization step With expectation values in hand for group membership, plug-in estimates are recomputed for the distribution parameters. The mixing coefficients ai are the means of the membership values over the N data points.

Section 5

Handwriting recognition The following example is based on an example in Christopher M. Bishop, Pattern Recognition and Machine Learning. Imagine that we are given an N×N black-and-white image that is known to be a scan of a hand-written digit between 0 and 9, but we don't know which digit is written. We can create a mixture model with K = 10 {\displaystyle K=10} different components, where each component is a vector of size N 2 {\displaystyle N^{2}} of Bernoulli distributions (one per pixel). Such a model can be trained with the expectation-maximization algorithm on an unlabeled set of hand-written digits, and will effectively cluster the images according to the digit being written. The same model could then be used to recognize the digit of another image simply by holding the parameters constant, computing the probability of the new image for each possible digit (a trivial calculation), and returning the digit that generated the highest probability.

Section 6

h s (j) (t) = w s (j) p s (x (t) ; μ s (j) , Σ s (j)) ∑ i = 1 n w i (j) p i (x (t) ; μ i (j) , Σ i (j)) . {\displaystyle h_{s}^{(j)}(t)={\frac {w_{s}^{(j)}p_{s}(x^{(t)};\mu _{s}^{(j)},\Sigma _{s}^{(j)})}{\sum _{i=1}^{n}w_{i}^{(j)}p_{i}(x^{(t)};\mu _{i}^{(j)},\Sigma _{i}^{(j)})}}.}

Section 7

The expectation step With initial guesses for the parameters of our mixture model, "partial membership" of each data point in each constituent distribution is computed by calculating expectation values for the membership variables of each data point. That is, for each data point xj and distribution Yi, the membership value yi, j is:

Section 8

Expectation maximization (EM) Expectation maximization (EM) is seemingly the most popular technique used to determine the parameters of a mixture with an a priori given number of components. This is a particular way of implementing maximum likelihood estimation for this problem. EM is of particular appeal for finite normal mixtures where closed-form expressions are possible such as in the following iterative algorithm by Dempster et al.

(1977)

Section 9

K = { p (·) : p (·) = ∑ i = 1 n a i f i (· ; θ i) , a i > 0 , ∑ i = 1 n a i = 1 , f i (· ; θ i) ∈ J ∀ i , n } {\displaystyle K=\left\{p(\cdot):p(\cdot)=\sum _{i=1}^{n}a_{i}f_{i}(\cdot ;\theta _{i}),a_{i}>0,\sum _{i=1}^{n}a_{i}=1,f_{i}(\cdot ;\theta _{i})\in J\ \forall i,n\right\}}

Section 10

History Mixture distributions and the problem of mixture decomposition, that is the identification of its constituent components and the parameters thereof, has been cited in the literature as far back as 1846 (Quetelet in McLachlan, 2000) although common reference is made to the work of Karl Pearson (1894) as the first author to explicitly address the decomposition problem in characterising non-normal attributes of forehead to body length ratios in female shore crab populations. The motivation for this work was provided by the zoologist Walter Frank Raphael Weldon who had speculated in 1893 (in Tarter and Lock) that asymmetry in the histogram of these ratios could signal evolutionary divergence. Pearson's approach was to fit a univariate mixture of two normals to the data by choosing the five parameters of the mixture such that the empirical moments matched that of the model. While his work was successful in identifying two potentially distinct sub-populations and in demonstrating the flexibility of mixtures as a moment matching tool, the formulation required the solution of a 9th degree (nonic) polynomial which at the time posed a significant computational challenge. Subsequent works focused on addressing these problems, but it was not until the advent of the modern computer and the popularisation of Maximum Likelihood (MLE) parameterisation techniques that research really took off. Since that time there has been a vast body of research on the subject spanning areas such as fisheries research, agriculture, botany, economics, medicine, genetics, psychology, palaeontology, electrophoresis, finance, geology and zoology.